Volume and Surface Area

1. Find the volume of a cube whose edge is 5 cm.

A) 100 cm^3

B) 125 cm^3

C) 150 cm^3

D) 25 cm^3

Answer: B) 125 cm^3

2. The length, breadth, and height of a cuboid are 10 cm, 8 cm, and 6 cm respectively. What is its volume?

A) 480 cm^3

B) 240 cm^3

C) 360 cm^3

D) 500 cm^3

Answer: A) 480 cm^3

3. What is the total surface area of a cube with a side length of 4 cm?

A) 64 cm^2

B) 96 cm^2

C) 16 cm^2

D) 36 cm^2

Answer: B) 96 cm^2

4. The radius of the base of a cylinder is 7 cm and its height is 10 cm. Find its volume. (Use $\pi = 22/7$)

A) 1540 cm^3

B) 1450 cm^3

C) 1500 cm^3

D) 1600 cm^3

Answer: A) 1540 cm^3

5. Find the volume of a cone with radius 3 cm and height 7 cm.

A) 44 cm^3

B) 66 cm^3

C) 33 cm^3

D) 22 cm^3

Answer: B) 66 cm^3

6. The surface area of a sphere is 616 cm^2. Find its radius.

A) 7 cm

B) 14 cm

C) 21 cm

D) 28 cm

Answer: A) 7 cm

7. A cuboid measures 15 m × 12 m × 10 m. Find its total surface area.

A) 900 m^2

B) 1800 m^2

C) 450 m^2

D) 950 m^2

Answer: A) 900 m^2

8. Two cubes each of volume 64 cm^3 are joined end to end. What is the surface area of the resulting cuboid?

A) 160 cm^2

B) 144 cm^2

C) 128 cm^2

D) 192 cm^2

Answer: A) 160 cm^2

9. The volume of a sphere is $(4/3)\pi r^3$. If the radius is 3 cm, the volume is:

A) 36π cm^3

B) $27\pi\ cm^3$

C) $18\pi\ cm^3$

D) $12\pi\ cm^3$

Answer: A) $36\pi\ cm^3$

10. A room is 12 m long, 8 m broad, and 4 m high. The length of the longest rod that can be placed in the room is:

A) 12 m

B) 14.4 m

C) 15 m

D) 16 m

Answer: B) 14.4 m

11. If the diagonal of a cube is $\sqrt{12}$ cm, what is its volume?

A) $8\ cm^3$

B) $12\ cm^3$

C) $24\ cm^3$

D) $64\ cm^3$

Answer: A) $8\ cm^3$

12. The ratio of the volumes of two cubes is 8:27. What is the ratio of their surface areas?

A) 2:3

B) 4:9

C) 8:27

D) 16:81

Answer: B) 4:9

13. How many bricks, each measuring 25 cm × 11.25 cm × 6 cm, will be needed to build a wall of 8 m × 6 m × 22.5 cm?

A) 5600

B) 6000

C) 6400

D) 7200

Answer: C) 6400

14. The curved surface area of a cylinder is 264 m^2 and its volume is 924 m^3. Find the ratio of its diameter to its height.

A) 3:7

B) 7:3

C) 6:7

D) 7:6

Answer: D) 7:6

15. Find the capacity of a tank of dimensions 8 m × 6 m × 2.5 m in liters.

A) 12000 liters

B) 120000 liters

C) 1200 liters

D) 120 liters

Answer: B) 120000 liters

16. If the radius of a sphere is doubled, its surface area becomes:

A) 2 times

B) 3 times

C) 4 times

D) 8 times

Answer: C) 4 times

17. A cone of height 24 cm has a curved surface area of 550 cm^2. Find its volume.

A) 1232 cm^3

B) 1200 cm^3

C) 1100 cm^3

D) 1000 cm^3

Answer: A) 1232 cm^3

18. A hemispherical bowl has a radius of 3.5 cm. What is the volume of water it can contain?

A) 89.8 cm^3

B) 44.9 cm^3

C) 179.6 cm^3

D) 90.5 cm^3

Answer: A) 89.8 cm^3

19. Three metal cubes with edges 6 cm, 8 cm, and 10 cm are melted to form a single new cube. The edge of the new cube is:

A) 12 cm

B) 14 cm

C) 16 cm

D) 18 cm

Answer: A) 12 cm

20. The volume of a wall is 5 times as high as it is broad and 8 times as long as it is high. If the volume is 12.8 m^3, find the breadth of the wall.

A) 40 cm

B) 45 cm

C) 50 cm

D) 55 cm

Answer: A) 40 cm

21. A cylindrical tank has a capacity of 6160 m^3. If the radius of its base is 14 m, find the depth of the tank.

A) 5 m

B) 10 m

C) 15 m

D) 20 m

Answer: B) 10 m

22. The surface area of a cube is 150 cm^2. Its volume is:

A) 64 cm^3

B) 125 cm^3

C) 216 cm^3

D) 343 cm^3

Answer: B) 125 cm^3

23. If the volume of a cube is 729 cm^3, what is the length of its diagonal?

A) $9\sqrt{2}$ cm

B) $9\sqrt{3}$ cm

C) 9 cm

D) 18 cm

Answer: B) $9\sqrt{3}$ cm

24. A hollow iron pipe is 21 cm long and its external diameter is 8 cm. If the thickness of the pipe is 1 cm and iron weighs 8 g/cm^3, the weight of the pipe is:

A) 3.696 kg

B) 3.6 kg

C) 36 kg

D) 36.9 kg

Answer: A) 3.696 kg

25. The volume of a right circular cone is 1232 cm^3 and its vertical height is 24 cm. Its curved surface area is:

A) 154 cm^2

B) 550 cm^2

C) 616 cm^2

D) 704 cm^2

Answer: B) 550 cm^2

26. Find the volume of the largest right circular cone that can be cut out of a cube of edge 9 cm.

A) 190.93 cm^3

B) 200.5 cm^3

C) 180.5 cm^3

D) 210 cm^3

Answer: A) 190.93 cm^3

27. A rectangular water tank is 80 m long and 25 m broad. Water flows into it through a pipe of 25 cm^2 cross-section at the rate of 16 km/hr. How much will the level of water rise in 45 minutes?

A) 2 cm

B) 2.5 cm

C) 1.5 cm

D) 3 cm

Answer: A) 2 cm

28. The curved surface area of a hemisphere is 2772 cm^2. Find its volume.

A) 19404 cm^3

B) 38808 cm^3

C) 9702 cm^3

D) 18000 cm^3

Answer: A) 19404 cm^3

29. If the height of a cylinder is doubled and the radius is halved, the volume will:

A) Double

B) Halve

C) Remain same

D) Become 4 times

Answer: B) Halve

30. A cone, a hemisphere, and a cylinder stand on equal bases and have the same height. The ratio of their volumes is:

A) 1 : 2 : 3

B) 2 : 1 : 3

C) 1 : 3 : 2

D) 3 : 1 : 2

Answer: A) 1 : 2 : 3

31. The lateral surface area of a cube is 256 m^2. The volume of the cube is:

A) 512 m^3

B) 64 m^3

C) 216 m^3

D) 256 m^3

Answer: A) 512 m^3

32. Find the number of lead balls, each 1 cm in diameter, that can be made from a sphere of diameter 12 cm.

A) 1728

B) 144

C) 1000

D) 512

Answer: A) 1728

33. A solid cylinder has a total surface area of 462 cm^2. Its curved surface area is one-third of its total surface area. Find the volume of the cylinder.

A) 530 cm^3

B) 539 cm^3

C) 545 cm^3

D) 550 cm^3

Answer: B) 539 cm^3

34. The diagonal of a rectangle is $\sqrt{41}$ cm and its area is 20 cm^2. The perimeter of the rectangle must be:

A) 9 cm

B) 18 cm

C) 20 cm

D) 22 cm

Answer: B) 18 cm

35. The radii of two cylinders are in the ratio 2:3 and their heights are in the ratio 5:3. The ratio of their volumes is:

A) 27:20

B) 20:27

C) 4:9

D) 9:4

Answer: B) 20:27

36. 50 men took a dip in a water tank 40 m long and 20 m broad. If the average displacement of water by a man is 4 m^3, then the rise in the water level in the tank will be:

A) 20 cm

B) 25 cm

C) 30 cm

D) 35 cm

Answer: B) 25 cm

37. A sphere and a cube have equal surface areas. The ratio of the volume of the sphere to that of the cube is:

A) $\sqrt{\pi} : \sqrt{6}$

B) $\sqrt{6} : \sqrt{\pi}$

C) $\sqrt{\pi} : \sqrt{3}$

D) $\sqrt{3} : \sqrt{\pi}$

Answer: B) $\sqrt{6} : \sqrt{\pi}$

38. The slant height of a conical mountain is 2.5 km and the area of its base is 1.54 km^2. Find the height of the mountain.

A) 2.2 km

B) 2.3 km

C) 2.4 km

D) 2.1 km

Answer: C) 2.4 km

39. The volume of a cuboid is twice the volume of a cube. If the dimensions of the cuboid are 9 cm, 8 cm, and 6 cm, the total surface area of the cube is:

A) 72 cm^2

B) 108 cm^2

C) 216 cm^2

D) 432 cm^2

Answer: C) 216 cm^2

40. How many meters of cloth 5 m wide will be required to make a conical tent, the radius of whose base is 7 m and height is 24 m?

A) 110 m

B) 120 m

C) 130 m

D) 100 m

Answer: A) 110 m

41. The sum of the length, breadth, and depth of a cuboid is 19 cm and the length of its diagonal is 11 cm. Find the surface area of the cuboid.

A) 240 cm^2

B) 250 cm^2

C) 260 cm^2

D) 270 cm^2

Answer: A) 240 cm^2

42. Two cones have their heights in the ratio 1:3 and the radii of their bases in the ratio 3:1. The ratio of their volumes is:

A) 1:1

B) 1:3

C) 3:1

D) 9:1

Answer: C) 3:1

43. A metallic sheet is of rectangular shape with dimensions 48 m × 36 m. From each one of its corners, a square is cut off so as to make an open box. If the length of the square is 8 m, the volume of the box is:

A) 4800 m^3

B) 5120 m^3

C) 6400 m^3

D) 8960 m^3

Answer: B) 5120 m^3

44. A right triangle with sides 3 cm, 4 cm, and 5 cm is rotated about the side of 3 cm to form a cone. The volume of the cone so formed is:

A) 12π cm^3

B) 15π cm^3

C) 16π cm^3

D) 20π cm^3

Answer: C) 16π cm^3

45. Determine the volume of a frustum of a cone if the radii of its ends are 10 cm and 4 cm and its height is 15 cm.

A) 2100π cm^3

B) 2280π cm^3

C) 780π cm^3

D) 1560π cm^3

Answer: D) 1560π cm^3

46. If the length of a diagonal of a cube is 6 cm, then its volume is:

A) $18\sqrt{3}$ cm^3

B) $24\sqrt{3}$ cm^3

C) $28\sqrt{3}$ cm^3

D) $30\sqrt{3}$ cm^3

Answer: B) $24\sqrt{3}$ cm^3

47. A large cube is formed from the material obtained by melting three smaller cubes of 3 cm, 4 cm and 5 cm side. What is the ratio of the total surface areas of the smaller cubes and the large cube?

A) 2:1

B) 3:2

C) 25:18

D) 27:20

Answer: C) 25:18

48. A cistern 6 m long and 4 m wide contains water up to a depth of 1 m 25 cm. The total area of the wet surface is:

A) 49 m^2

B) 50 m^2

C) 53.5 m^2

D) 55 m^2

Answer: A) 49 m^2

49. A cylindrical bucket 28 cm in diameter and 72 cm high is full of water. The water is emptied into a rectangular tank 66 cm long and 28 cm wide. Find the height of the water level in the tank.

A) 20 cm

B) 24 cm

C) 28 cm

D) 32 cm

Answer: B) 24 cm

50. If the volume of two cubes are in the ratio 27:64, the ratio of their edges is:

A) 3:4

B) 3:8

C) 9:16

D) 4:3

Answer: A) 3:4

51. The length of the longest pole that can be kept in a room 5 m long, 4 m broad and 3 m high is:

A) $5\sqrt{2}$ m

B) $6\sqrt{2}$ m

C) $7\sqrt{2}$ m

D) $8\sqrt{2}$ m

Answer: A) $5\sqrt{2}$ m

52. What is the total surface area of a solid hemisphere of radius 10 cm? (Use $\pi = 3.14$)

A) 942 cm^2

B) 900 cm^2

C) 628 cm^2

D) 314 cm^2

Answer: A) 942 cm^2

53. The height of a cylinder is 14 cm and its curved surface area is 264 cm^2. The volume of the cylinder is:

A) 308 cm^3

B) 396 cm^3

C) 412 cm^3

D) 450 cm^3

Answer: B) 396 cm^3

54. A copper sphere of diameter 18 cm is drawn into a wire of diameter 4 mm. The length of the wire is:

A) 2.43 m

B) 24.3 m

C) 243 m

D) 2430 m

Answer: C) 243 m

55. If the total surface area of a cube is 96 cm^2, its volume is:

A) 16 cm^3

B) 32 cm^3

C) 64 cm^3

D) 128 cm^3

Answer: C) 64 cm^3

56. The radius of a cylinder is 10 cm and height is 4 cm. The number of centimeters that may be added either to the radius or to the height to get the same increase in the volume of the cylinder is:

A) 4 cm

B) 5 cm

C) 16 cm

D) 20 cm

Answer: B) 5 cm

57. A conical tent is to accommodate 10 persons. Each person must have 6 m^2 of space to sit and 30 m^3 of air to breathe. What will be the height of the cone?

A) 10 m

B) 15 m

C) 20 m

D) 25 m

Answer: B) 15 m

58. Water flows at the rate of 10 meters per minute from a cylindrical pipe 5 mm in diameter. How long would it take to fill a conical vessel whose diameter at the base is 40 cm and depth 24 cm?

A) 48 min 15 sec

B) 51 min 12 sec

C) 52 min 1 sec

D) 55 min

Answer: B) 51 min 12 sec

59. If each edge of a cube is increased by 50%, the percentage increase in its surface area is:

A) 50%

B) 75%

C) 100%

D) 125%

Answer: D) 125%

60. The number of bricks, each measuring 24 cm × 12 cm × 8 cm, required to construct a wall 24 m long, 8 m high and 60 cm thick, if 10% of the wall is filled with mortar is:

A) 35000

B) 40000

C) 45000

D) 50000

Answer: C) 45000

61. A solid metallic sphere of radius 8 cm is melted and recast into spherical balls each of radius 2 cm. The number of such balls is:

A) 32

B) 64

C) 16

D) 48

Answer: B) 64

62. A cylindrical rod of iron whose height is 8 times its radius is melted and cast into spherical balls each of half the radius of the cylinder. The number of such spherical balls is:

A) 12

B) 16

C) 24

D) 48

Answer: D) 48

63. The areas of three adjacent faces of a cuboid are x, y and z. If the volume is V, then V^2 is equal to:

A) xy

B) yz

C) zx

D) xyz

Answer: D) xyz

64. If the radius of the base of a cone is halved, keeping the height same, the ratio of the volume of the reduced cone to that of the original cone is:

A) 1:2

B) 1:4

C) 1:8

D) 1:16

Answer: B) 1:4

65. A cylinder and a cone have equal radii of their bases and equal heights. If their curved surface areas are in the ratio 8:5, the ratio of their radius and height is:

A) 3:4

B) 3:5

C) 4:3

D) 5:4

Answer: A) 3:4

66. The volume of a sphere is divided by its surface area. The result is 27 cm. The radius of the sphere is:

A) 9 cm

B) 27 cm

C) 54 cm

D) 81 cm

Answer: D) 81 cm

67. A well with 10 m inside diameter is dug 14 m deep. Earth taken out of it is spread all around to a width of 5 m to form an embankment. Find the height of the embankment.

A) 2.53 m

B) 3.53 m

C) 4.66 m

D) 5.12 m

Answer: C) 4.66 m

68. The surface area of a solid sphere is increased by 21% without changing its shape. Find the percentage increase in its volume.

A) 30%

B) 33.1%

C) 42%

D) 44%

Answer: B) 33.1%

69. Water is flowing at the rate of 5 km/hr through a pipe of diameter 14 cm into a rectangular tank which is 50 m long and 44 m wide. Determine the time in which the level of the water in the tank will rise by 7 cm.

A) 1 hour

B) 1.5 hours

C) 2 hours

D) 2.5 hours

Answer: C) 2 hours

70. A rectangular block 6 cm by 12 cm by 15 cm is cut up into an exact number of equal cubes. The least possible number of cubes will be:

A) 6

B) 11

C) 33

D) 40

Answer: D) 40

71. A cube of side 6 cm is cut into a number of cubes each of side 2 cm. The number of cubes will be:

A) 6

B) 9

C) 12

D) 27

Answer: D) 27

72. If the areas of three adjacent faces of a rectangular box are 120 cm^2, 72 cm^2 and 60 cm^2, find the volume of the box.

A) 720 cm^3

B) 800 cm^3

C) 850 cm^3

D) 900 cm^3

Answer: A) 720 cm^3

73. The paint in a certain container is sufficient to paint an area equal to 9.375 m^2. How many bricks of dimensions 22.5 cm × 10 cm × 7.5 cm can be painted out of this container?

A) 100

B) 200

C) 500

D) 1000

Answer: A) 100

74. A river 3 m deep and 40 m wide is flowing at the rate of 2 km per hour. How much water will fall into the sea in a minute?

A) 4000 m^3

B) 40000 m^3

C) 400 m^3

D) 2000 m^3

Answer: A) 4000 m^3

75. The height of a cone is 30 cm. A small cone is cut off at the top by a plane parallel to the base. If its volume be 1/27 of the volume of the given cone, at what height above the base is the section made?

A) 10 cm

B) 15 cm

C) 20 cm

D) 25 cm

Answer: C) 20 cm

76. The curved surface area of a cylindrical pillar is 264 m^2 and its volume is 924 m^3. Find the diameter and height of the pillar.

A) 14 m, 6 m

B) 12 m, 7 m

C) 7 m, 12 m

D) 14 m, 7 m

Answer: A) 14 m, 6 m

77. Find the weight of a solid rectangular iron piece of size 50 cm × 40 cm × 10 cm, if 1 cm^3 of iron weighs 8 gm.

A) 160 kg

B) 150 kg

C) 140 kg

D) 180 kg

Answer: A) 160 kg

78. The radius of a sphere is increased by 10%. The volume will increase by:

A) 20%

B) 30%

C) 33.1%

D) 10%

Answer: C) 33.1%

79. Three solid spheres of gold whose radii are 1 cm, 6 cm and 8 cm respectively are melted into a single solid sphere. Find the radius of the sphere.

A) 9 cm

B) 10 cm

C) 12 cm

D) 14 cm

Answer: A) 9 cm

80. A cone and a hemisphere have equal bases and equal volumes. The ratio of their heights is:

A) 1:2

B) 2:1

C) 3:1

D) 1:3

Answer: B) 2:1

81. A cylinder is circumscribed about a hemisphere and a cone is inscribed in the cylinder so as to have its vertex at the center of one end and the other end as its base. The volume of the cylinder, hemisphere and cone are, respectively in the ratio:

A) 2:3:1

B) 3:2:1

C) 3:1:2

D) 1:2:3

Answer: B) 3:2:1

82. A cuboid has total surface area of 40 m^2 and its lateral surface area is 26 m^2. Find the area of its base.

A) 7 m^2

B) 9 m^2

C) 14 m^2

D) 20 m^2

Answer: A) 7 m^2

83. The sum of the radius of the base and the height of a solid cylinder is 37 m. If the total surface area of the solid cylinder is 1628 m^2, find the volume of the cylinder.

A) 4620 m^3

B) 4260 m^3

C) 6420 m^3

D) 2460 m^3

Answer: A) 4620 m^3

84. The radius and height of a right circular cone are in the ratio 5:12. If its volume is 314 m^3, find the slant height and the radius. (Use $\pi = 3.14$)

A) 13 m, 5 m

B) 12 m, 5 m

C) 13 m, 12 m

D) 26 m, 10 m

Answer: A) 13 m, 5 m

85. A vessel is in the form of a hollow hemisphere mounted by a hollow cylinder. The diameter of the hemisphere is 14 cm and the total height of the vessel is 13 cm. Find the inner surface area of the vessel.

A) 572 cm^2

B) 500 cm^2

C) 472 cm^2

D) 612 cm^2

Answer: A) 572 cm^2

86. A toy is in the form of a cone of radius 3.5 cm mounted on a hemisphere of same radius. The total height of the toy is 15.5 cm. Find the total surface area of the toy.

A) 214.5 cm^2

B) 230 cm^2

C) 200 cm^2

D) 210 cm^2

Answer: A) 214.5 cm^2

87. How many silver coins, 1.75 cm in diameter and of thickness 2 mm, must be melted to form a cuboid of dimensions 5.5 cm × 10 cm × 3.5 cm?

A) 200

B) 300

C) 400

D) 500

Answer: C) 400

88. A metallic sphere of radius 4.2 cm is melted and recast into the shape of a cylinder of radius 6 cm. Find the height of the cylinder.

A) 2.4 cm

B) 2.74 cm

C) 3.0 cm

D) 3.5 cm

Answer: B) 2.74 cm

89. A 20 m deep well with diameter 7 m is dug and the earth from digging is evenly spread out to form a platform 22 m by 14 m. Find the height of the platform.

A) 2.5 m

B) 3 m

C) 3.5 m

D) 4 m

Answer: A) 2.5 m

90. A container shaped like a right circular cylinder having diameter 12 cm and height 15 cm is full of ice cream. The ice cream is to be filled into cones of height 12 cm and diameter 6 cm, having a hemispherical shape on the top. The number of such cones which can be filled with ice cream is:

A) 5

B) 8

C) 10

D) 15

Answer: C) 10

91. The volume of the largest right circular cone that can be cut out of a cube of edge 7 cm is:

A) 89.8 cm^3

B) 90.5 cm^3

C) 100.2 cm^3

D) 78.6 cm^3

Answer: A) 89.8 cm^3

92. Two cubes have their volumes in the ratio 1:27. The ratio of the area of the face of one to that of the other is:

A) 1:3

B) 1:6

C) 1:9

D) 1:18

Answer: C) 1:9

93. A cube of edge 15 cm is immersed completely in a rectangular vessel containing water. If the dimensions of the base of vessel are 20 cm × 15 cm, find the rise in water level.

A) 9 cm

B) 10.25 cm

C) 11.25 cm

D) 12 cm

Answer: C) 11.25 cm

94. The slant height of a frustum of a cone is 4 cm and the perimeters (circumference) of its circular ends are 18 cm and 6 cm. Find the curved surface area of the frustum.

A) 48 cm^2

B) 50 cm^2

C) 52 cm^2

D) 54 cm^2

Answer: A) 48 cm^2

95. A solid iron rectangular block of dimensions 4.4 m, 2.6 m and 1 m is cast into a hollow cylindrical pipe of internal radius 30 cm and thickness 5 cm. Find the length of the pipe.

A) 110 m

B) 112 m

C) 115 m

D) 120 m

Answer: B) 112 m

96. If the radius of a cylinder is decreased by 50% and height is increased by 50%, the volume decreases by:

A) 50.5%

B) 60%

C) 62.5%

D) 65%

Answer: C) 62.5%

97. Water flows in a tank 150 m × 100 m at the base through a pipe whose cross-section is 2 dm by 1.5 dm at the speed of 15 km/hr. In what time will the water be 3 meters deep?

A) 50 hours

B) 100 hours

C) 150 hours

D) 200 hours

Answer: B) 100 hours

98. The surface area of a sphere is same as the curved surface area of a right circular cylinder whose height and diameter are 12 cm each. The radius of the sphere is:

A) 3 cm

B) 4 cm

C) 6 cm

D) 12 cm

Answer: C) 6 cm

99. A circus tent is cylindrical to a height of 3 m and conical above it. If its diameter is 105 m and the slant height of the conical portion is 53 m, calculate the length of the canvas 5 m wide to make the required tent.

A) 1900 m

B) 1947 m

C) 1980 m

D) 2000 m

Answer: B) 1947 m

100. A sphere of radius 6 cm is dropped into a cylindrical vessel partly filled with water. The radius of the vessel is 8 cm. If the sphere is submerged completely, then the surface of the water rises by:

A) 2.25 cm

B) 2.50 cm

C) 3.75 cm

D) 4.50 cm

Answer: A) 2.25 cm